PULSE Command Center

What is the state of the marketplace, in four sentences?

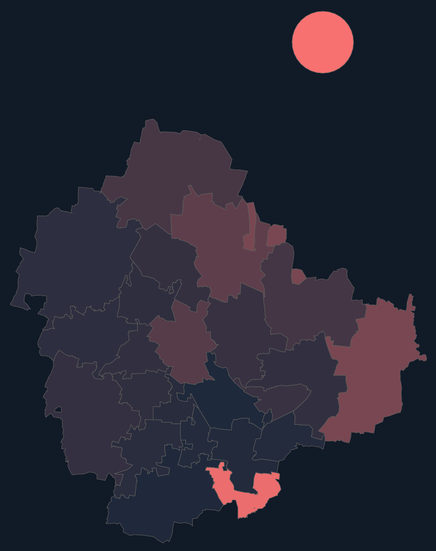
service: Food Delivery | Mobility

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Command Center

- Command Center
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- Demand Map
- Supply & Idle Partners
- Pressure Index
- Forecast & Early Warning
- Policy Simulator
- Repositioning Flows
- Incentives & Stress Tests
- Decision Playbook

9,811	71.9%	52.8%	935	₹3,37,205
Demand per Day	Ride Completion %	Idle %	Lost Jobs per Day	Revenue per Day

Jobs lost to no partner, per day



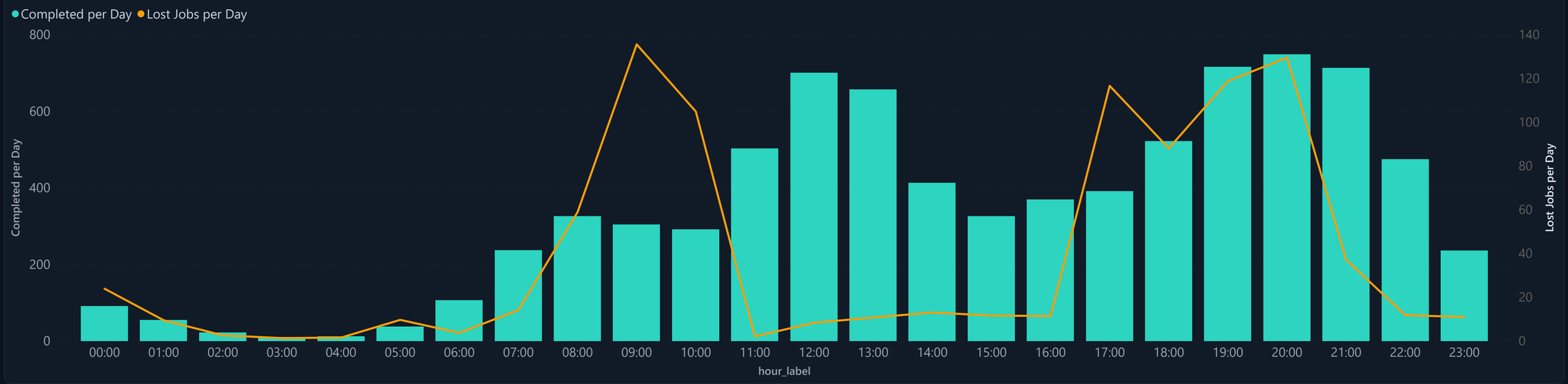
Partners sit idle 53% of their online time, yet 28% of ride requests are not completed.

A week-ahead forecast flags 29% of zone-hours - and they hold 91% of the jobs later lost.

Profit repositioning cuts lost jobs by 13.0% for +₹205 a day - a service lever, not a profit lever.

Bonuses recover jobs at ₹43 each; they pay only below ₹16/hour on a normal day, ₹26 on festival evenings.

A day in Bengaluru: jobs completed vs lost, per hour



Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5

Marketplace Performance

Which services, zones and hours fail customers?

period: All

day_type: All

zone_type: All

service: Food Delivery | Mobility

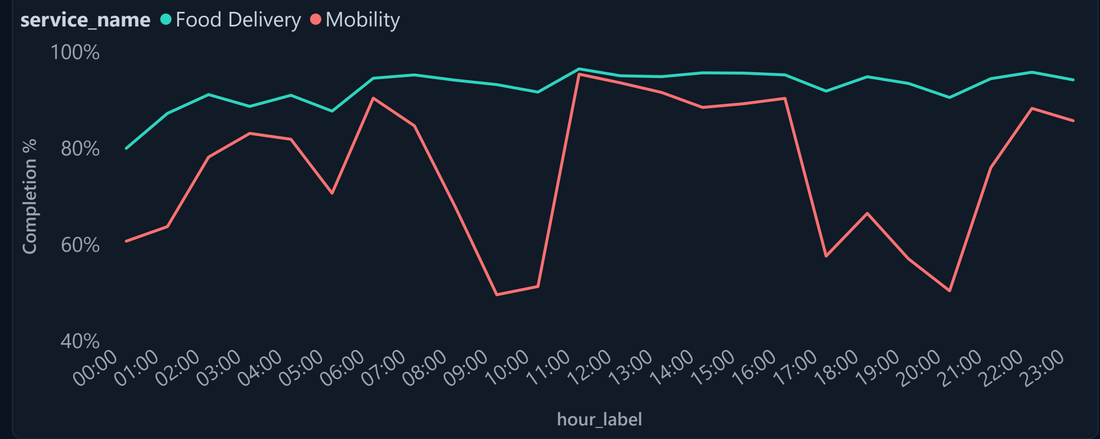
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84.3%	9.5%	4.5%	8.9	29.2	₹40.79	13.6%
Completion %	Lost to No Partner %	Customer Cancel %	Avg Pickup ETA (min)	Avg Order-to-Door (min)	Revenue per Completed Job	Take Rate %

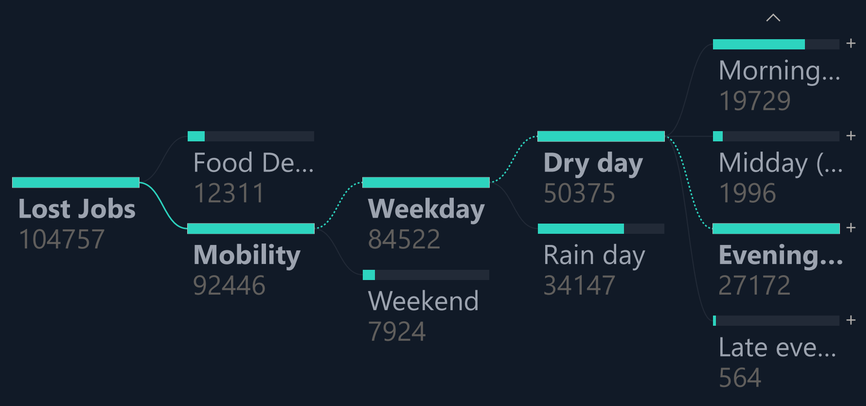
zone_type	Demand per Day	Completion %	Lost to No Partner %	Revenue per Day
⊞ Office	1,620	71.5%	22.5%	₹47,491
⊞ Transit hub	766	64.7%	24.4%	₹13,771
⊞ Restaurant cluster	1,765	91.4%	3.1%	₹70,511
⊞ Mixed	2,103	86.9%	6.7%	₹73,564
⊞ Residential	3,557	89.2%	5.3%	₹1,31,868
Total	**9,811**	**84.3%**	**9.5%**	**₹3,37,205**

Completion by hour: rides fail in the evening peak



Where do lost jobs come from? Click + to drill

service... × Mobility | day_type × Weekday | weather × Dry day | daypart ×



Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5

Demand Map

Where and when does demand arrive?

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Metric

Jobs lost per day	Demand per day	Completion %	Partners idle %	Pressure index (MPI)	Value of one more partner-hour

service

Food Delivery	Mobility

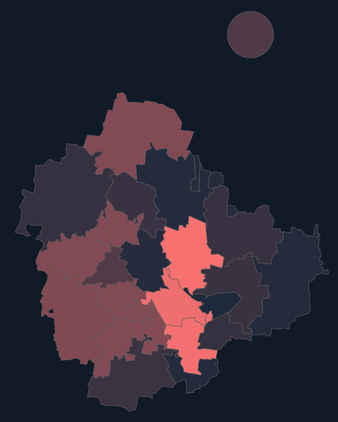
Extra demand on rain days

22.8%

Hours shown on the map

hour_of_day: 0 – 23

Bengaluru - Demand per day by zone

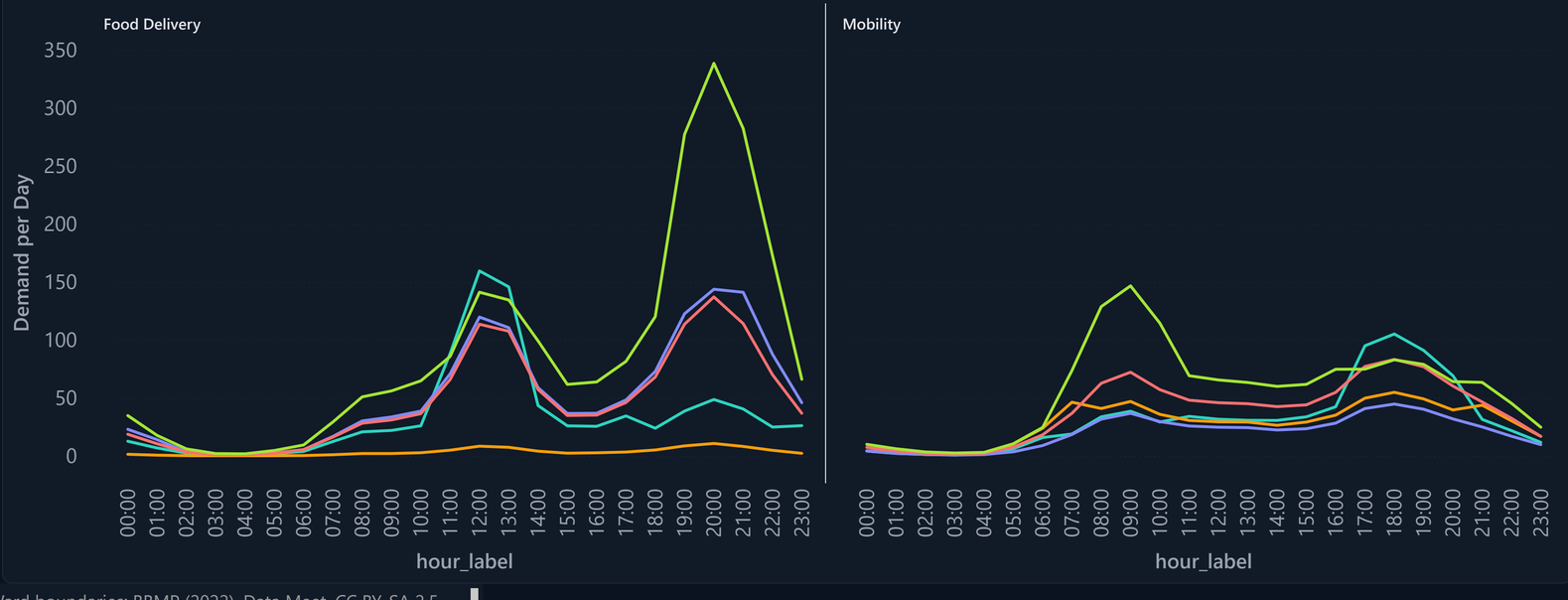


zone_label × hour_of_day

zone_label	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
BEL - Bellandur-ORR																								
BGR - Bannerghatta Road																								
BSK - Banashankari																								
BTM - BTM Layout																								
BVG - Basavanagudi																								
CBD - CBD (MG Road)																								
ECY - Electronic City																								
HEB - Hebbal																								
HSR - HSR Layout																								
IND - Indiranagar																								
JAY - Jayanagar																								
JPN - JP Nagar																								
KIA - Kempegowda Airport																								
KOR - Koramangala																								
KRP - KR Puram																								
MAJ - Majestic																								
MAN - Manyata Tech Park																								
MAR - Marathahalli																								
MLM - Malleshwaram																								
RAJ - Rajajinagar																								
SAR - Sarjapur Road																								
WHF - Whitefield																								
YEL - Yelahanka																								
YPR - Yeshwanthpur																								

Hourly demand by zone type: rides peak with the commute, food at mealtimes

zone_type: Office, Transit hub, Restaurant cluster, Mixed, Residential



Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5

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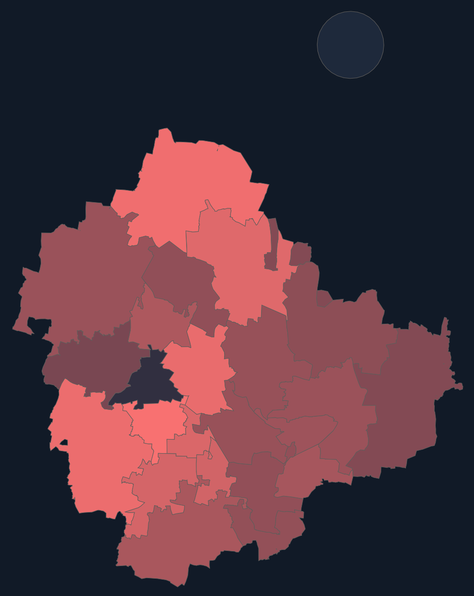
Supply & Idle Partners

Where are partners, and why are they idle?

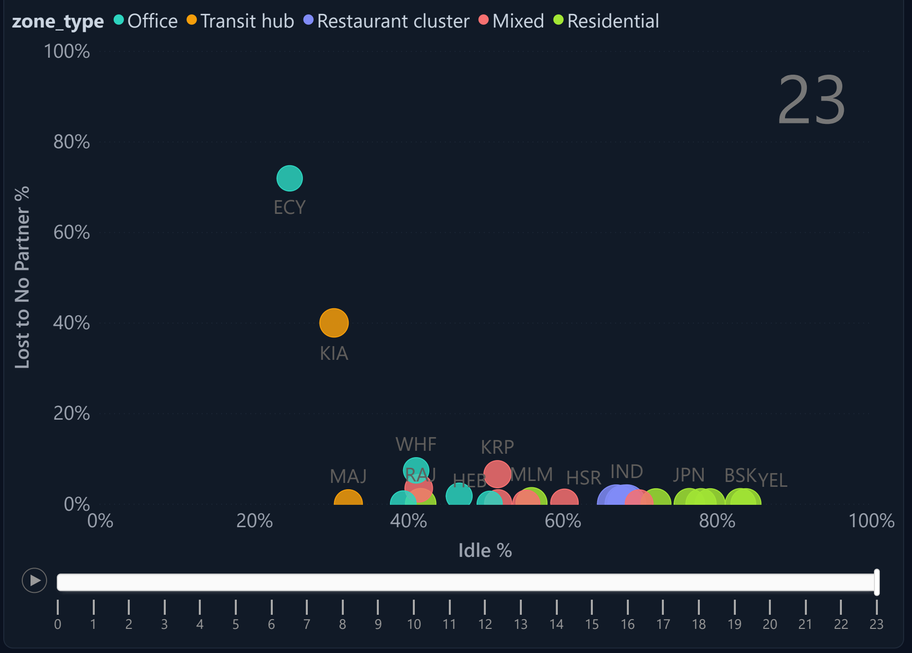
service: Food Delivery | Mobility

239.0	47.2%	52.8%	3,027	9.2
Avg Partners Online	Utilization %	Idle %	Idle Partner-Hours per Day	Empty km per Busy Hour

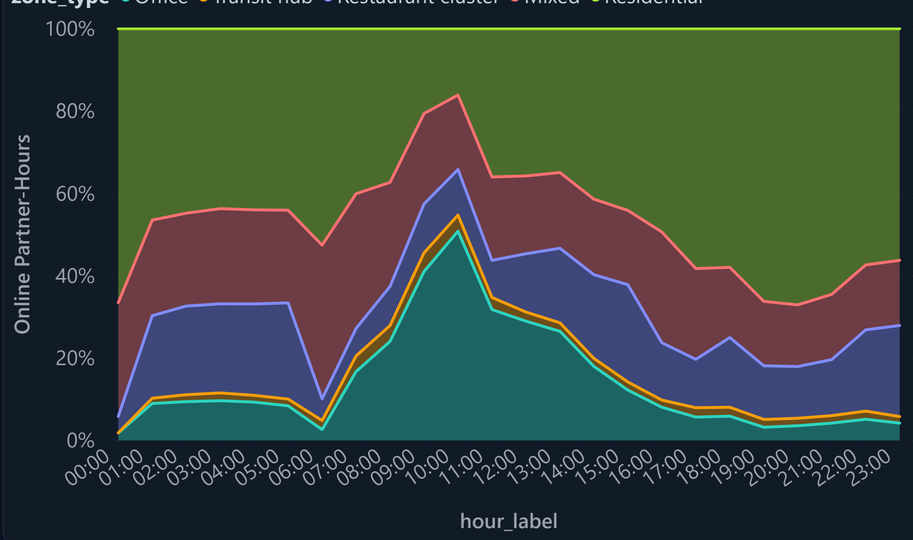
Share of partner time spent idle, by zone



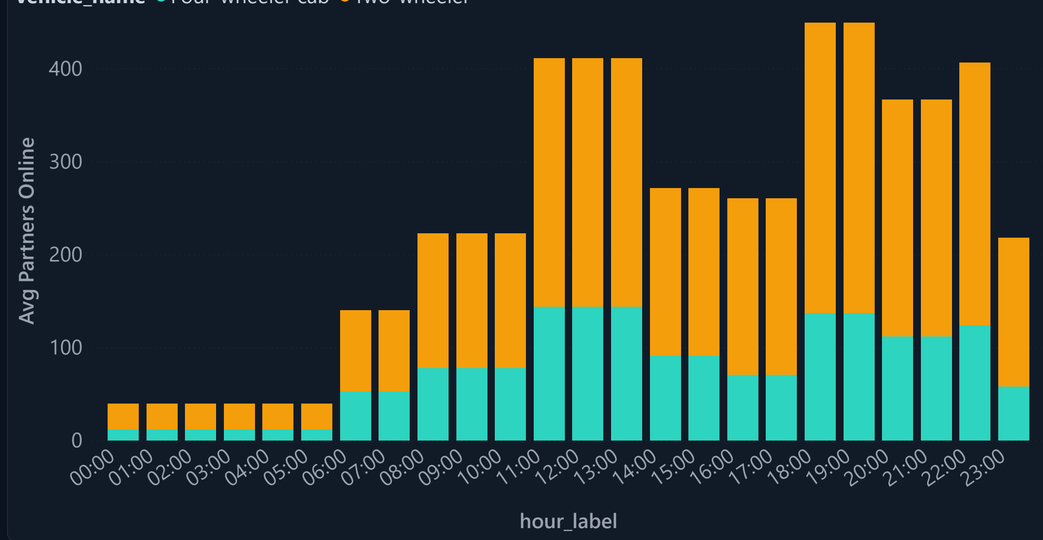
Press play: idle partners and lost jobs are in different zones at the same hour



Where partners are, by hour (share of all partners online)



Partners online by hour and vehicle



Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5

Pressure Index

Where is demand above supply, and what kind of shortage is it?

service: Food Delivery | Mobility

Day type: Weekday | Weekend

PULSE
Command Center

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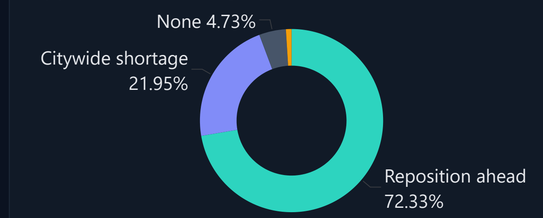
0.69 City MPI

33.6% Under-supplied Zone-Hours %

72.3% Reposition-Ahead Share of Lost %

22.0% Citywide Share of Lost %

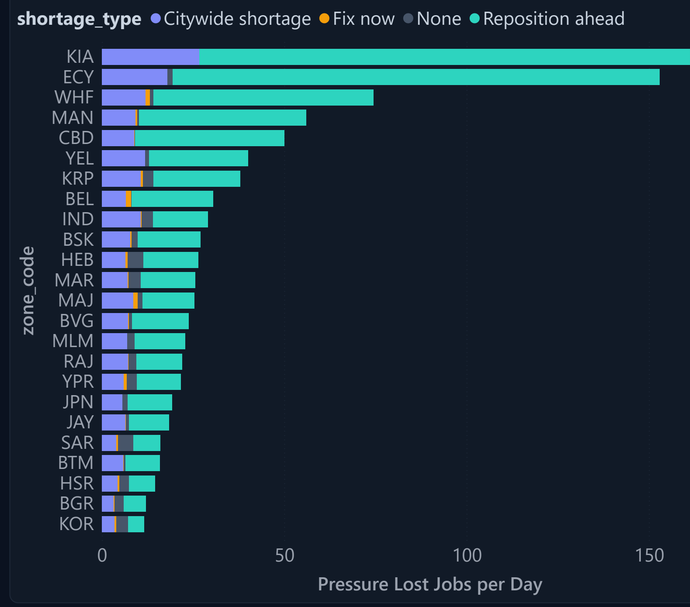
Why jobs are lost: shortage types



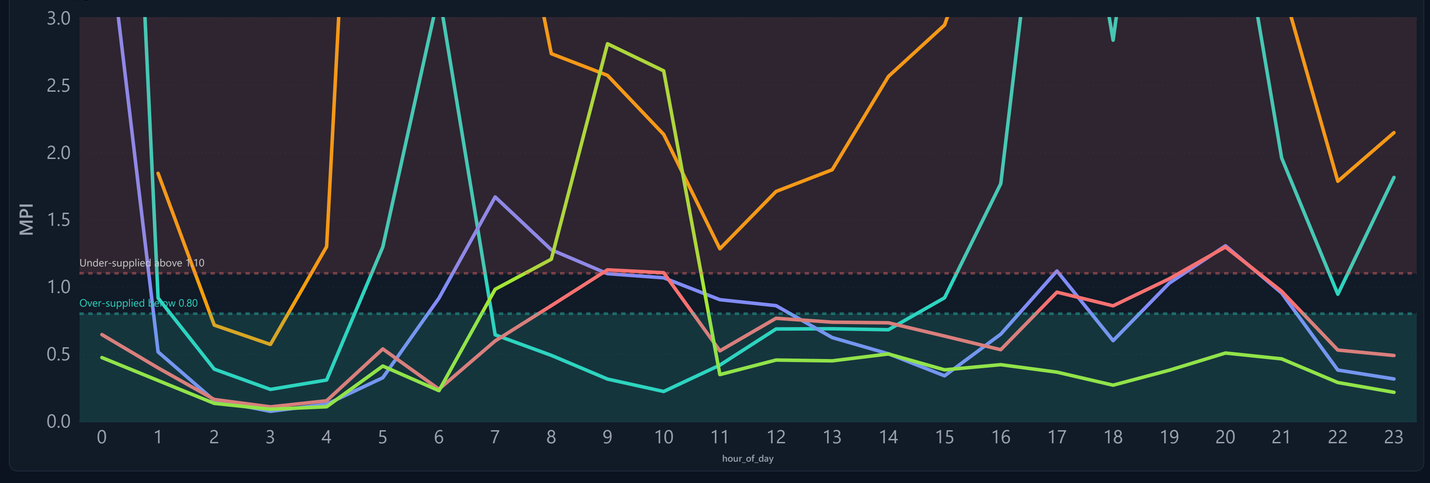
Pressure index, weekdays (above 1.10 = under-supplied)

zone_label	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
BEL - Bellandur-ORR	1.84	0.64	0.28	0.16	0.15	1.13	2.99	0.60	0.48	0.30	0.23	0.49	0.71	0.70	0.60	1.04	2.03	5.04	2.14	4.53	3.14	1.37	0.70	1.74
BGR - Bannerghatta Road	0.81	0.43	0.17	0.11	0.15	0.48	0.33	0.69	0.92	1.30	1.31	0.58	0.86	0.73	0.66	0.63	0.46	0.80	0.75	0.94	1.07	0.77	0.45	0.43
BSK - Banashankari	0.54	0.28	0.12	0.09	0.10	0.37	0.28	1.13	1.08	2.87	2.83	0.41	0.50	0.48	0.54	0.40	0.38	0.32	0.25	0.36	0.44	0.41	0.23	0.16
BTM - BTM Layout	1.07	0.41	0.18	0.09	0.16	0.54	0.29	1.34	1.20	2.60	2.21	0.30	0.42	0.41	0.42	0.34	0.38	0.35	0.25	0.33	0.45	0.42	0.27	0.23
BVG - Basavanagudi	0.40	0.33	0.14	0.09	0.09	0.38	0.24	0.80	1.65	2.74	2.35	0.29	0.33	0.33	0.34	0.27	0.41	0.34	0.19	0.26	0.35	0.32	0.21	0.17
CBD - CBD (MG Road)		0.66	0.29	0.17	0.26	1.04	3.49	0.37	0.31	0.18	0.12	0.23	0.37	0.37	0.42	0.54	1.12	3.17	1.96	4.96	3.50	1.47	0.73	1.31
ECY - Electronic City		31.58	6.17	4.17	8.17	19.04	3.22	1.64	1.04	0.65	0.41	0.69	1.44	1.72	2.29	2.51	2.35	16.88	11.19	13.31	13.78	7.72	3.61	5.57
HEB - Hebbal	1.69	0.72	0.26	0.16	0.28	0.89	0.24	0.56	0.93	0.94	0.87	0.60	0.82	0.78	0.84	0.63	0.57	1.33	1.07	1.41	1.48	1.16	0.60	0.47
HSR - HSR Layout	1.31	0.45	0.12	0.06	0.12	0.29	1.43	2.03	1.08	0.98	1.07	0.88	0.86	0.61	0.50	0.35	0.77	1.28	0.63	1.27	1.30	0.97	0.39	0.33
IND - Indiranagar		0.55	0.15	0.06	0.12	0.34	1.08	1.64	1.35	1.20	1.03	0.88	0.87	0.63	0.53	0.34	0.59	1.12	0.65	1.08	1.42	1.05	0.37	0.32
JAY - Jayanagar		0.50	0.20	0.09	0.16	0.60	0.28	1.07	1.44	2.45	2.22	0.53	0.63	0.56	0.68	0.40	0.35	0.28	0.21	0.29	0.42	0.39	0.24	0.18
JPN - JP Nagar	0.25	0.31	0.16	0.08	0.08	0.42	0.21	0.89	1.05	2.97	2.58	0.34	0.47	0.46	0.51	0.39	0.54	0.39	0.27	0.37	0.52	0.43	0.28	0.21
KIA - Kempegowda Airport					32.50	248.66	3.05	6.19	6.22	6.50	4.60	1.75	1.86	1.97	3.15	3.77	2.91	4.83	3.75	3.63	3.40	3.43	2.25	2.11
KOR - Koramangala		0.57	0.21	0.10	0.14	0.34	0.63	1.42	1.46	1.13	1.10	0.96	0.85	0.62	0.48	0.32	0.60	0.99	0.53	0.82	1.21	0.86	0.38	0.29
KRP - KR Puram	0.52	0.31	0.15	0.06	0.10	0.43	0.33	0.83	1.48	1.91	1.48	0.63	0.88	0.81	0.80	0.75	0.71	1.09	0.86	0.96	1.32	1.09	0.59	0.59
MAJ - Majestic		1.03	0.33	0.27	0.63	4.64	8.98	3.97	1.75	1.63	1.39	1.02	1.59	1.78	2.15	2.41	7.03	4.52	2.65	2.71	2.88	3.11	1.49	2.19
MAN - Manyata Tech Park		1.03	0.38	0.36	0.30	1.16	1.49	0.48	0.33	0.25	0.19	0.33	0.58	0.59	0.61	0.81	1.48	5.18	2.85	5.24	4.27	1.89	0.84	1.48
MAR - Marathahalli	0.52	0.49	0.27	0.16	0.25	0.86	0.13	0.41	0.54	0.92	1.12	0.34	0.64	0.78	0.83	0.64	0.63	1.23	1.03	1.00	1.32	1.04	0.58	0.67
MLM - Malleshwaram	0.36	0.28	0.12	0.12	0.17	0.51	0.17	0.74	1.16	2.67	2.28	0.28	0.39	0.40	0.47	0.34	0.37	0.35	0.30	0.46	0.71	0.72	0.49	0.35
RAJ - Rajajinagar	0.42	0.39	0.19	0.13	0.15	0.69	0.24	2.58	1.04	2.64	2.64	0.29	0.44	0.50	0.63	0.59	0.52	0.63	0.50	0.80	0.99	0.84	0.57	0.47
SAR - Sarjapur Road	0.33	0.20	0.07	0.07	0.10	0.29	0.21	0.55	0.77	0.97	0.86	0.43	0.57	0.54	0.58	0.47	0.59	1.21	0.85	1.08	1.24	0.82	0.44	0.38
WHF - Whitefield		0.70	0.35	0.15	0.25	0.93		1.16	0.97	0.71	0.50	0.90	1.35	1.12	0.85	1.06	2.74	5.69	2.91	6.23	4.78	2.17	0.96	1.81
YEL - Yelahanka	0.38	0.16	0.06	0.05	0.06	0.20	0.18	0.69	1.24	3.90	5.49	0.49	0.60	0.56	0.60	0.48	0.49	0.42	0.33	0.48	0.60	0.52	0.28	0.18
YPR - Yeshwanthpur	1.71	0.68	0.20	0.17	0.18	0.83	0.37	0.78	0.97	1.16	1.23	0.83	1.01	0.86	0.77	0.77	0.38	0.58	0.70	1.09	1.44	1.08	0.56	0.50

Jobs lost per day, by zone and shortage type



When each zone type crosses the shortage line (pressure index by hour)



Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5

Forecast & Early Warning

can we see shortages a week ahead?

service: Food Delivery | Mobility

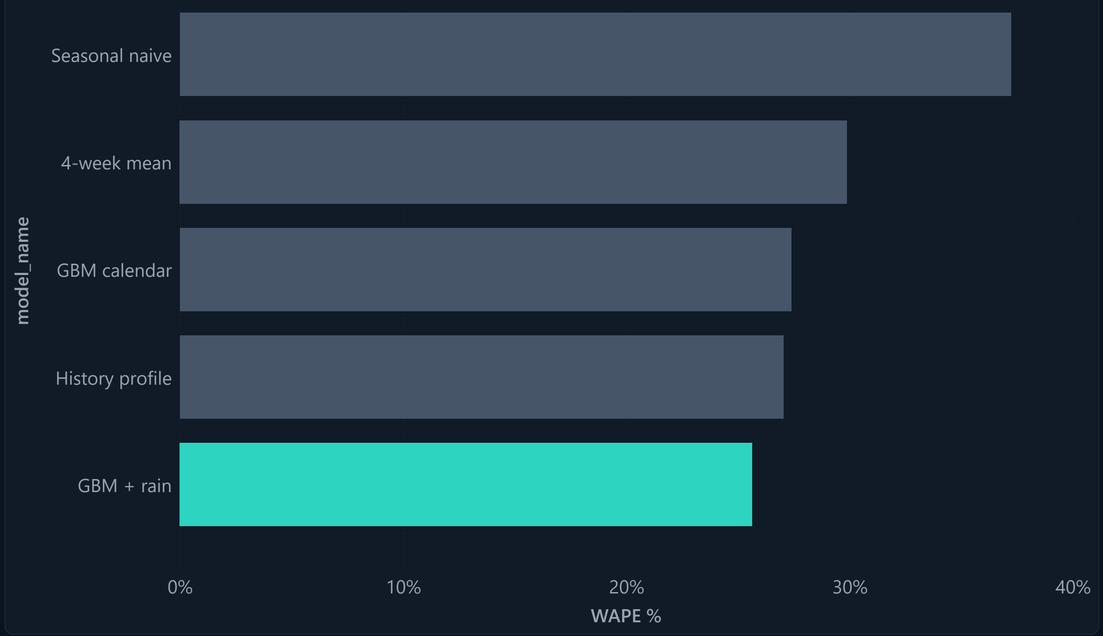
Forecast model: GBM + rain

PULSE
Command Center

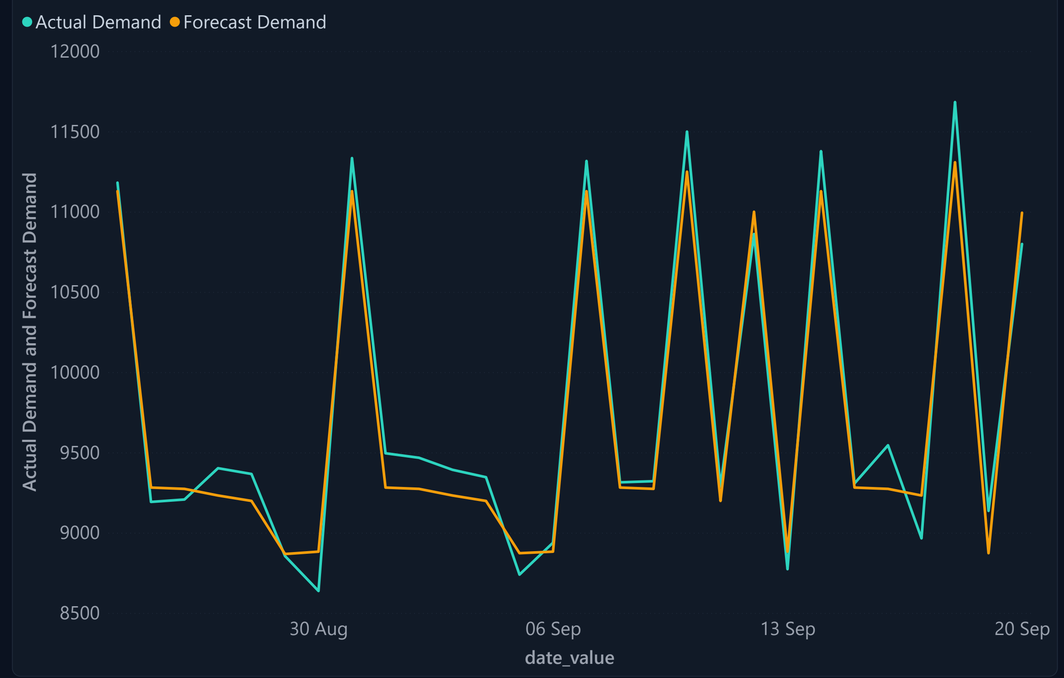
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25.6%	-0.7%	31.1%	28.8%	80.1%	69.6%	90.7%
WAPE %	Forecast Bias %	Skill vs Seasonal Naive %	Flag Rate %	Precision %	Recall %	Lost Jobs Captured %

Forecast error by model (lower is better)



Holdout weeks: actual demand vs week-ahead forecast



Week-ahead warnings vs what actually happened

outcome	Warning Zone-Hours	Warning Lost Jobs	Share of Warning Lost Jobs %
Correct warning	3,724	23,501	88.4%
Missed shortage	1,628	1,924	7.2%
False alarm	923	612	2.3%
Correct all-clear	9,853	537	2.0%
Total	**16,128**	**26,574**	**100.0%**

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Policy Simulator

Which policy wins, and at what value of a customer?

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What is one completed job worth to the business? (₹)

20

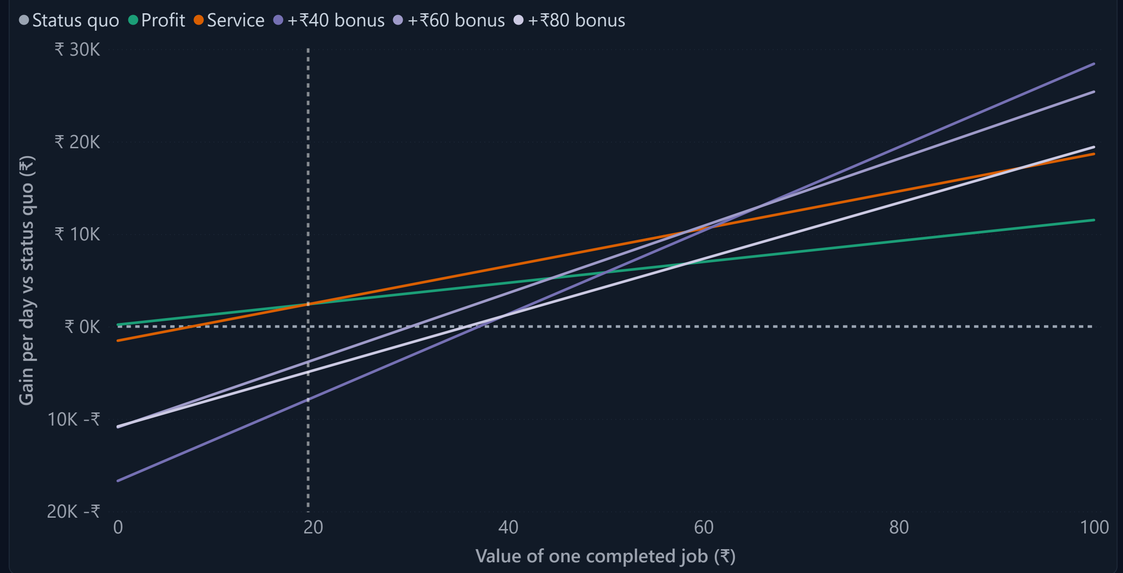
Policy

Status quo	Profit	Service	Perfect forecast	+₹40 bonus	+₹60 bonus	+₹80 bonus

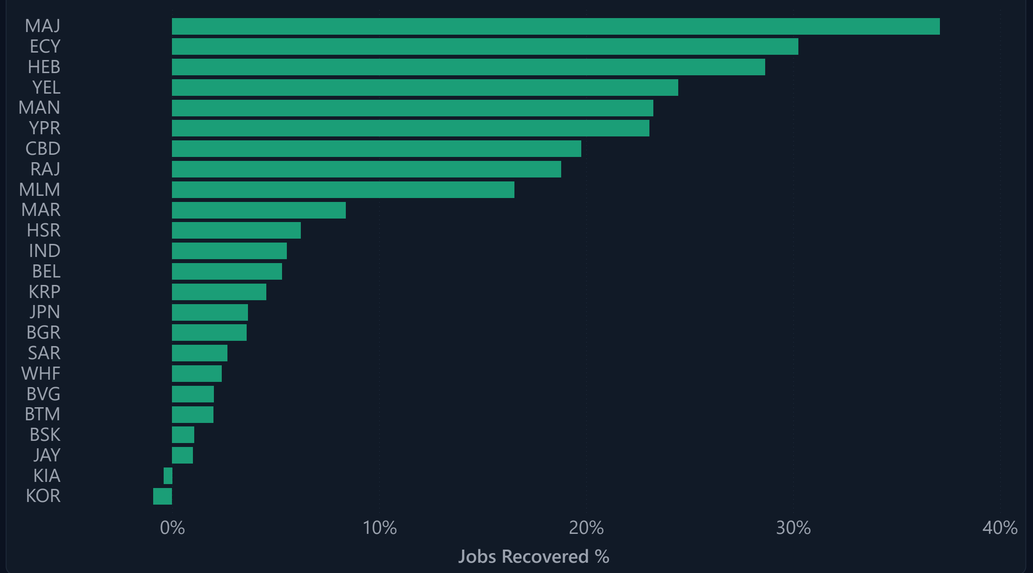
Optimizer (service, ₹20 goodwill)

Gain per day vs status quo **₹2.510K**

Each line is a policy. Where lines cross, the better policy changes



Share of lost jobs won back, by zone



scenario_name	Contribution per Day	Contribution Uplift per Day	Econ Lost Jobs vs SQ %	Econ Extra Completed per Day	Econ Moves per Day	Break-even Job Value
Status quo	₹ 3,36,968	₹0	0.0%	0		
Optimizer (profit)	₹ 3,37,173	₹205	-13.0%	113	86	
Optimizer (service, ₹20 goodwill)	₹ 3,35,438	-₹1,530	-23.9%	202	178	₹ 19.5
Optimizer, perfect forecast	₹ 3,37,406	₹438	-13.3%	118	87	
Profit + ₹40 bonus	₹ 3,20,268	-₹16,700	-49.5%	451		₹ 50.0
Profit + ₹60 bonus	₹ 3,26,073	-₹10,895	-40.3%	363		₹ 44.4
Profit + ₹80 bonus	₹ 3,26,162	-₹10,806	-34.1%	302		₹ 58.2

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Repositioning Flows

Who moves where, and what is one more partner worth?

Policy

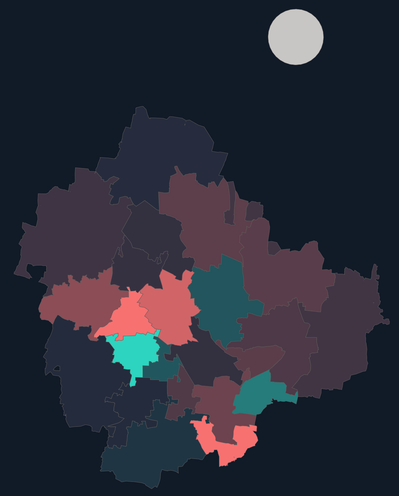
- Profit
- Service

85.9
Moves per Day

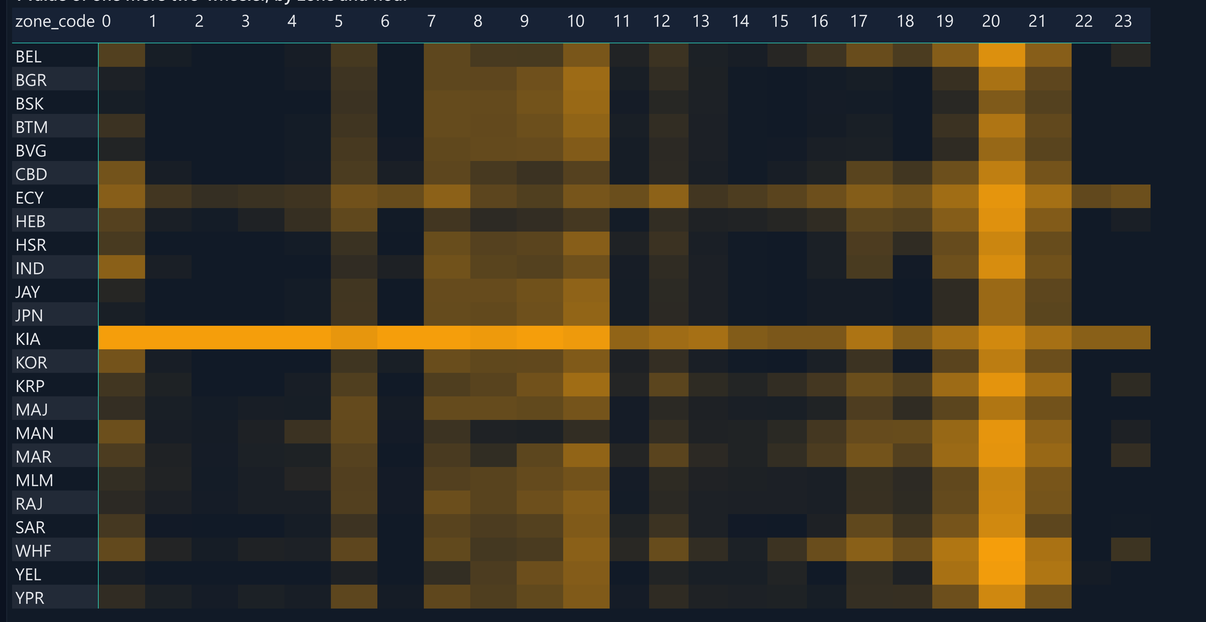
7.55
Avg Move km

₹3,891
Move Cost per Day

Net partners moved in per day (red = receives, teal = sends)



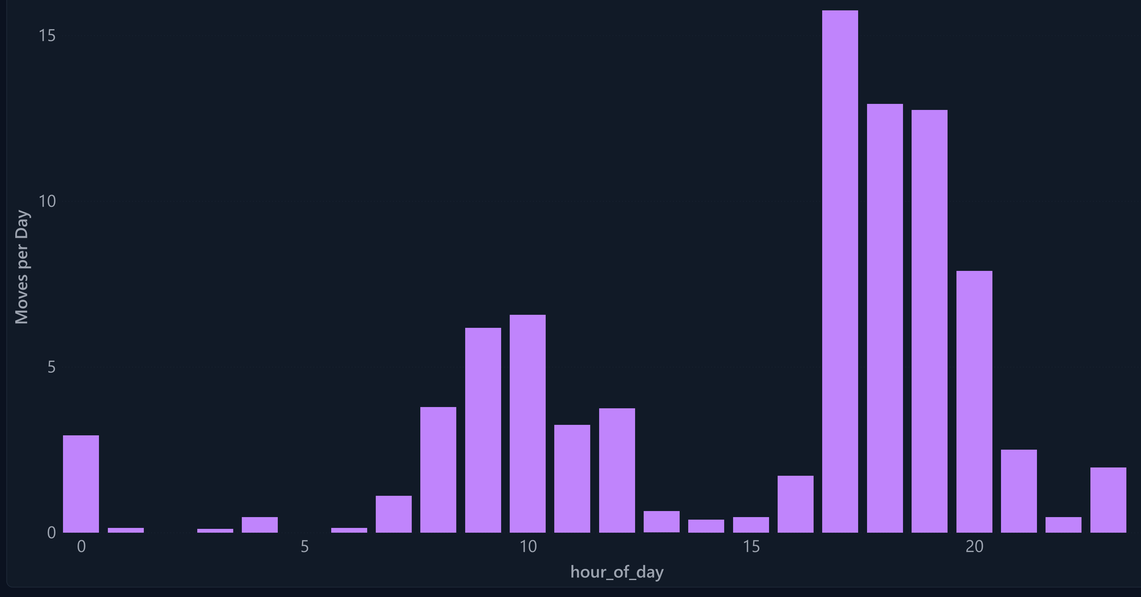
₹ value of one more two-wheeler, by zone and hour



Who moves where: partners per day, from (rows) → to (columns)

zone_code	BEL	BGR	BSK	BTM	BVG	CBD	ECY	HEB	HSR	IND	JAY	JPN	KOR	KRP	MAJ	MAN	MAR	RAJ	SAR	WHF	YEL	YPR
BEL																			0.0		0.1	
BGR						4.0		0.5			0.1											
BSK				0.3															1.5			
BTM	0.1							2.1	0.1		0.7											
BVG		0.0			5.3				0.4	0.1		7.6						1.9				
CBD						0.0			0.5	0.4	0.1	3.5	0.2									
ECY		0.2						0.5													0.1	
HEB									0.0					0.1							0.5	0.1
HSR				0.1		0.5					0.2				0.1							
IND					6.0							1.5			0.3							
JAY				3.7	0.2	1.6		0.3	0.5	0.0		0.1										
JPN		1.3		0.1				0.1		0.3												
KOR	1.3					0.1		0.3														
KRP									0.3												0.1	
MAJ																			0.1			
MAN									0.1		0.1											7.5
MAR	0.2								0.1				0.4							0.2	0.2	
MLM						0.6		1.6		0.3						0.2		1.3				0.1
RAJ				0.0										0.0								
SAR	0.2						7.5												1.3			
WHF													0.1						0.1			
YEL																9.0						
YPR																						

Moves by hour: partners are sent ahead of the evening peak



Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5 India.

Incentives & Stress Tests

When does paying partners pay?

PULSE
Command Center

- Command Center
- Marketplace Performance
- Demand Map
- Supply & Idle Partners
- Pressure Index
- Forecast & Early Warning
- Policy Simulator
- Repositioning Flows
- Incentives & Stress Tests
- Decision Playbook

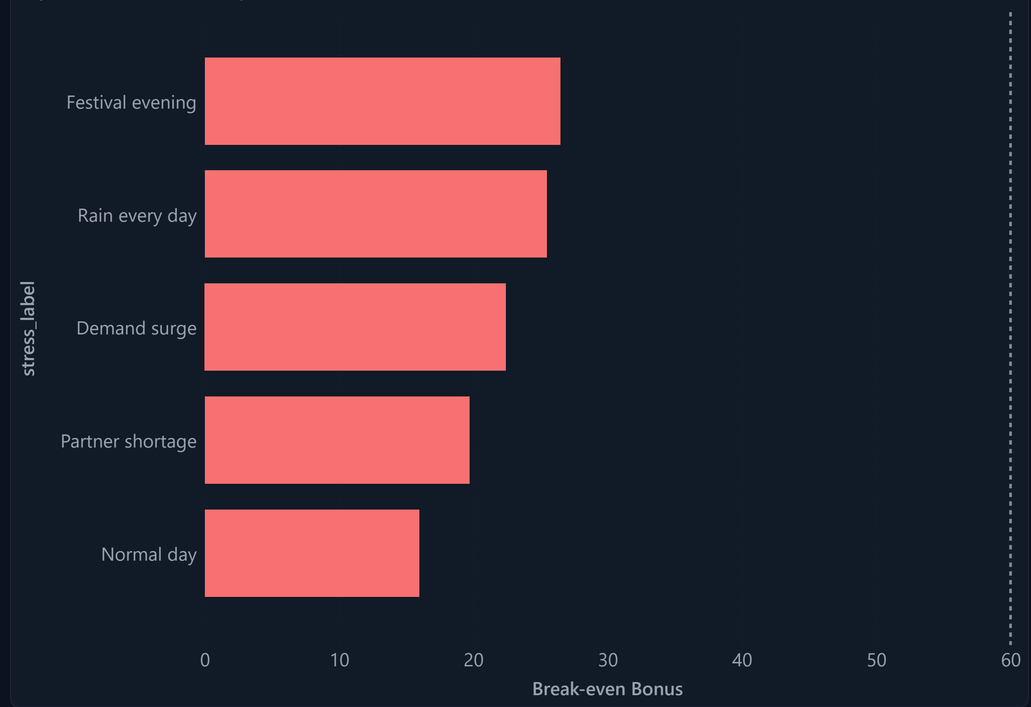
Situations (of 5) where this bonus pays

0

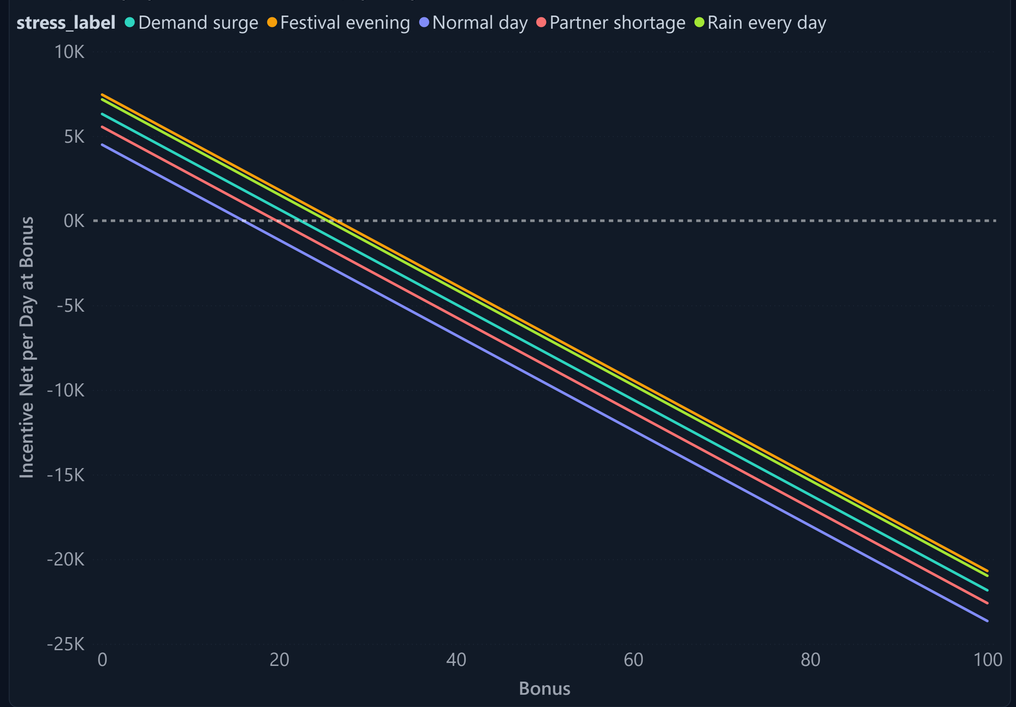
Bonus per guaranteed partner-hour (₹)

60

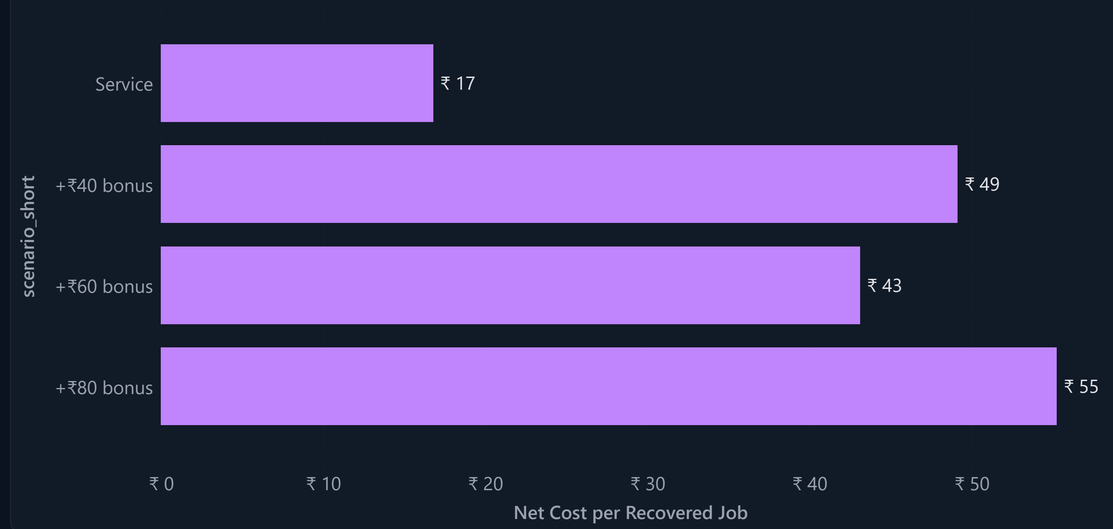
Highest bonus that still pays, per situation



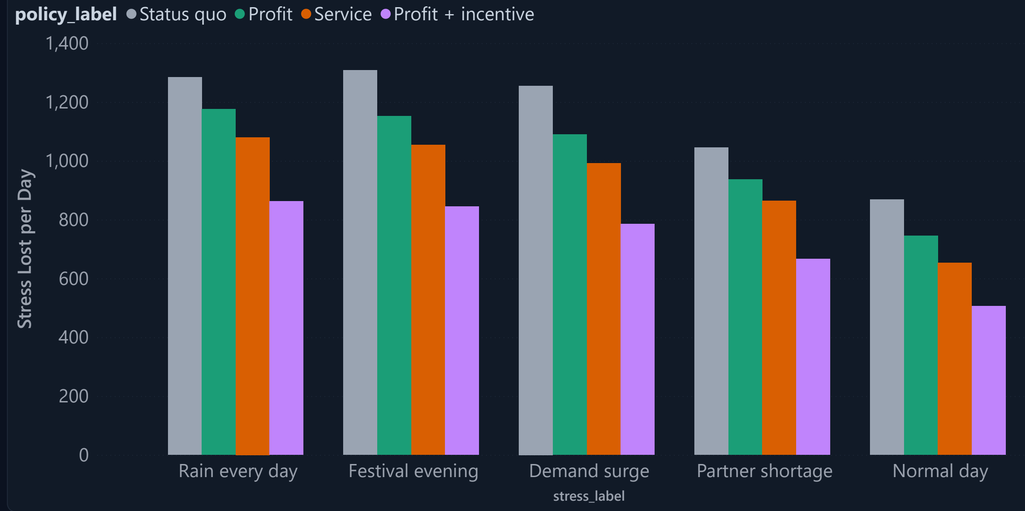
Net ₹ per day by bonus level: a bonus pays only above the line



Cost per job won back, vs profit repositioning



Jobs lost per day in each stress situation, by policy



Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5

PULSE
Command Center

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- Decision Playbook

Decision Playbook

What should each zone do on Monday?

Profit repositioning: gain per day
+₹205

Service setting: ₹ per job won back
₹ 16.8

₹60 bonus: ₹ per job won back
₹ 43.1

Zone playbook (sorted by jobs lost per day)

zone_label	zone_type	Lost Jobs per Day	Reposition-Ahead Share of Lost %	Citywide Share of Lost %	Jobs Recovered %	Zone Playbook
BEL - Bellandur-ORR	Office	30	73.2%	21.6%	5.3%	Repositioning plus same-hour dispatch
BGR - Bannerghatta Road	Mixed	12	49.7%	26.8%	3.6%	Repositioning plus same-hour dispatch
BSK - Banashankari	Residential	27	63.5%	29.1%	1.1%	Repositioning cannot fix it: station dedicated supply
BTM - BTM Layout	Residential	16	59.3%	36.9%	2.0%	Repositioning cannot fix it: station dedicated supply
BVG - Basavanagudi	Residential	24	65.1%	30.2%	2.0%	Repositioning cannot fix it: station dedicated supply
CBD - CBD (MG Road)	Office	50	81.7%	17.8%	19.8%	Repositioning plus same-hour dispatch
ECY - Electronic City	Office	153	87.3%	11.7%	30.2%	Pre-position partners before the peak
HEB - Hebbal	Mixed	26	57.0%	24.6%	28.6%	Pre-position partners before the peak
HSR - HSR Layout	Restaurant cluster	15	49.0%	29.9%	6.2%	Repositioning plus same-hour dispatch
IND - Indiranagar	Restaurant cluster	29	51.7%	36.6%	5.5%	Add partners: targeted bonus on rain and festival days
JAY - Jayanagar	Residential	18	59.4%	35.4%	1.0%	Service setting: profit policy skips this zone
JPN - JP Nagar	Residential	19	63.4%	29.4%	3.7%	Repositioning cannot fix it: station dedicated supply
KIA - Kempegowda Airport	Transit hub	162	83.5%	16.5%	-0.4%	Repositioning cannot fix it: station dedicated supply
KOR - Koramangala	Restaurant cluster	12	37.4%	30.8%	-0.9%	Add partners: targeted bonus on rain and festival days
KRP - KR Puram	Mixed	38	62.6%	28.1%	4.5%	Service setting: profit policy skips this zone
MAJ - Majestic	Transit hub	25	56.2%	34.2%	37.1%	Pre-position partners before the peak
MAN - Manyata Tech Park	Office	56	81.7%	16.5%	23.2%	Pre-position partners before the peak
MAR - Marathahalli	Mixed	26	58.4%	27.9%	8.4%	Repositioning plus same-hour dispatch
MLM - Malleshwaram	Residential	23	60.5%	30.4%	16.5%	Add partners: targeted bonus on rain and festival days
RAJ - Rajajinagar	Residential	22	56.8%	33.0%	18.8%	Add partners: targeted bonus on rain and festival days
SAR - Sarjapur Road	Mixed	16	46.1%	25.2%	2.7%	Repositioning plus same-hour dispatch
WHF - Whitefield	Office	74	81.0%	16.1%	2.4%	Repositioning cannot fix it: station dedicated supply
YEL - Yelahanka	Residential	40	67.7%	29.8%	24.5%	Pre-position partners before the peak
YPR - Yeshwanthpur	Mixed	22	55.7%	27.6%	23.1%	Pre-position partners before the peak

What to do on Monday

1. **Run profit repositioning every day.** It wins back about 13% of lost jobs at no extra cost.
2. **Switch to the service setting** if a completed job is worth more than about ₹20 (see Policy Simulator).
3. **Pre-position before the peak, not during it,** using the week-ahead warnings, which flag about 91% of lost jobs.
4. **Pay bonuses only on rain and festival days,** below about ₹25 an hour. Never as a standing programme.
5. **Station dedicated supply at the red zones,** starting with the airport (KIA).

Limitations: synthetic city; model-based estimates; one replay of four holdout weeks; partners assumed to follow every move

Synthetic Bengaluru — model-based estimates, not real-world impact. Ward boundaries: BBMP (2022), Data Meet, CC BY-SA 2.5